

StockSense

Complete Product & Technical Documentation

AI-Powered Stock Prediction & Portfolio Intelligence Platform

Indian & US Equity Markets | Nifty 100 | S&P 500 | Crypto

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Coverage	22 sections · All signals · Every factor

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1. Product Overview

StockSense is an AI-powered stock prediction and portfolio intelligence platform built for Indian and US equity markets. It combines institutional-grade quantitative signals with a consumer-friendly interface to deliver actionable BUY / HOLD / SELL signals with full explainability.

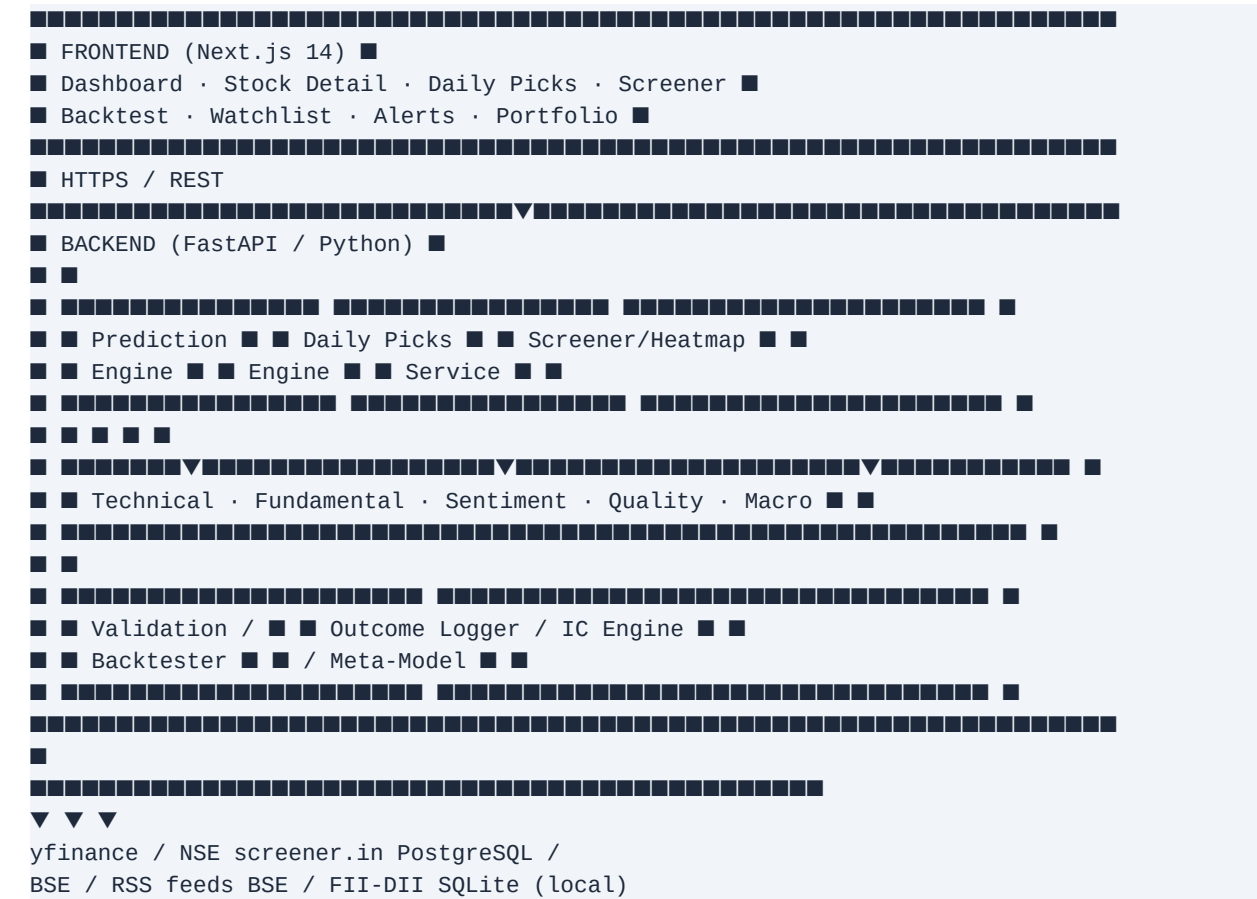
What StockSense Does

- Generates **BUY / HOLD / SELL signals** for Nifty 100, US large-cap, and top cryptocurrencies
- Delivers **Daily Picks** every morning at 9 AM IST — top 5 BUY ideas per horizon (short / medium / long)
- Shows **why** every signal was generated — factor breakdown, confidence scores, reasoning bullets
- Provides **trade levels** — entry zone, stop-loss, and target price with R:R ratio
- Runs a **learning engine** — tracks prediction outcomes and retrains factor weights weekly
- Supports **screener, backtest, watchlist, and alerts**

Supported Markets

Market	Universe	Horizons
India (NSE)	Nifty 100	Short (1–5 days), Medium (2–4 weeks), Long (3–6 months)
US	S&P 500 large-caps	Short, Medium, Long
Crypto	BTC, ETH, BNB, SOL, XRP, DOGE, ADA, SHIB, Matic, Doge	Short, Medium, Long

2. Architecture Overview



Tech Stack

Layer	Technology
Frontend	Next.js 14, React, TypeScript, TailwindCSS, Recharts
Backend	Python 3.11, FastAPI, Uvicorn
Auth	Supabase (JWT)
Database	PostgreSQL (production), SQLite (local)
Cache	In-memory (TTL-based), disk (picks_cache.json)
Hosting	Render.com (backend), Vercel (frontend)
Automation	GitHub Actions (cron jobs)

3. Data Sources

Source	Data Provided	Update Frequency	Used For
yfinance	Price, volume, P/E, ROE, margins, FCF, etc. (15–60 days)	Real-time (15–60 days)	All markets
screener.in	10-year financials, ROCE, 3Y/5Y CAGR, Patheki fields, promoter % pledge, cashflow	Daily (4h, 1h)	India, pledge %, cashflow
BSE API	Current fundamentals for renamed stocks	Daily (1h fallback)	India fallback
NSE FII/DII API	Daily institutional flows (■ Cr)	Daily (30min cache)	India quality signal
NSE Pledge API	Promoter pledge %, quarterly	Quarterly	India risk signal
Yahoo Finance RSS	News headlines per symbol	Real-time (10min cache)	Sentiment
Google News RSS	"{symbol} stock India" search	Real-time (10min cache)	Sentiment fallback
Economic Times RSS	India economy & stock news	Real-time	India sentiment
MoneyControl RSS	Stock & sector news	Real-time	India sentiment
yfinance macro tickers	S&P 500, VIX, Crude, Gold, US Dollar, Nikkei, DAX, etc.	Daily (15min, 1h)	Global macro

Data Fallback Chain (India)

yfinance → screener.in (if <5 key fields) → BSE API (if still sparse)

4. Core Prediction Engine

File: backend/services/prediction_engine.py

Signal Generation Formula

```
Composite Score = (Tech × W_tech) + (Fund × W_fund) + (Sent × W_sent)
+ Global Macro Adjustment
+ Analyst Consensus Adjustment
+ 52-Week Position Adjustment
+ Quality Factor Adjustment
+ Rounding Adjustment
- Risk Penalty
```

All component scores are on a **0–100 scale** (50 = neutral). The composite is also 0–100.

Signal Thresholds

Composite Score	Signal	Confidence Calculation
≥ 70	**BUY**	<code>`(score – 70) / 30 × 100%`</code>
55 – 69	**HOLD**	<code>`min(25%, abs(score – 62) × 3%)`</code>
< 55	**SELL**	<code>`(55 – score) / 55 × 100%`</code>

Dynamic Weights (Horizon × Volatility × Regime)

Base weights by horizon:

Horizon	Technical	Fundamental	Sentiment
Short (1–5 days)	70%	15%	15%
Medium (2–4 weeks)	40%	45%	15%
Long (3–6 months)	15%	75%	10%

Volatility modulation (applied on top of base weights):

Volatility Level	Annualised Vol	Technical	Fundamental
High	> 35%	Reduced to 10%	Boosted
Normal	15–35%	Base	Base
Low	< 15%	Boosted to 75%	Reduced

Regime modulation:

- **BULL regime:** Boost technical, reduce fundamental
- **BEAR regime:** Boost fundamental, reduce technical
- **SIDEWAYS:** No modulation (enables mean-reversion trades)

Prediction Caching

- **TTL:** 15 minutes per (symbol:market:horizon) key
- **Max size:** 300 entries (Render 512 MB free-tier limit)
- **Eviction:** LRU — oldest entry dropped when full
- **Async pattern:** First request returns HTTP 202 (computing); client polls every 5s; result cached on completion

5. Technical Analysis Module

File: backend/services/technical_indicators.py

Indicators Computed

Category	Indicator	Parameters
Momentum	RSI	14-period
	Stochastic RSI	14-period
	Williams %R	14-period
	CCI	20-period
	Stochastic Oscillator	K=14, D=3
Trend	MACD	12/26/9 EMA
	EMA	20, 50, 200 periods
	ADX + DI+/DI–	14-period

Category	Indicator	Parameters
Volatility	Bollinger Bands	20-period, 2σ
	ATR	14-period
Volume	OBV	Rolling
	Volume SMA	20-period
	VWAP	20-day rolling

Scoring Rules

RSI:

- < 30 (oversold): **+15 pts**
- $30-45$ (recovering): **+7 pts**
- $60-70$ (elevated): **-7 pts**
- > 70 (overbought): **-15 pts**

MACD:

- Above signal line: **+12 pts**
- Below signal line: **-12 pts**

EMA Trend:

- Price $>$ EMA200 (bull market structure): **+10 pts**
- EMA20 $>$ EMA50 (golden cross zone): **+8 pts**
- EMA20 $<$ EMA50 (death cross zone): **-8 pts**

ADX (Trend Strength):

- ADX > 25 AND $+DI > -DI$ (strong uptrend): **+10 pts**
- ADX > 25 AND $-DI > +DI$ (strong downtrend): **-10 pts**
- ADX ≤ 25 (sideways): **0 pts**

Bollinger Bands (%B):

- Near lower band (< 0.1): **+8 pts** (oversold)
- Near upper band (> 0.9): **-8 pts** (overbought)

Volume Confirmation:

- Up-day volume $>$ down-day volume: **+8 pts**
- Down-day volume $>$ up-day volume: **-8 pts**

Candlestick Patterns:

Pattern	Points	Signal
Hammer	+5	BUY
Morning Star	+5	BUY
Bullish Engulfing	+5	BUY
Shooting Star	-5	SELL
Evening Star	-5	SELL
Bearish Engulfing	-5	SELL

Technical Signal Output

- Score ≥ 58 : **BUY**
- Score ≤ 42 : **SELL**
- 42–58: **HOLD**

6. Fundamental Scoring Module

File: backend/services/prediction_engine.py → `_fundamental_score()`

Quality Gate (Hard Rejection)

A stock is flagged **REJECTED** before scoring if:

- ROE $< -10\%$ AND Profit Margin $< -5\%$
- Negative Operating Cash Flow (medium/long horizons only)
- D/E ratio $> 500\%$ (non-financial sector)

Scoring Components

Valuation

Metric	Threshold (India / US)	Points
P/E	< 18 IN / < 15 US (cheap)	+12
P/E	< 30 IN / < 25 US (fair)	+6
P/E	> 50 IN / > 40 US (expensive)	-10
P/B	< 2.5 IN / < 2 US	+6
PEG	< 0.75	+16
PEG	0.75–1.0	+10
PEG	> 2.5	-12
FCF Yield	$> 5\%$	+10
FCF Yield	3–5%	+5

Growth

Metric	Threshold	Points
Revenue Growth YoY	$> 20\%$	+12
Revenue Growth YoY	5–20%	+5
Revenue Growth YoY	$< -5\%$	-10
Revenue CAGR 3Y (screener)	$> 15\%$	+8
Revenue CAGR 3Y	$< 0\%$	-6
Profit CAGR 3Y (screener)	$> 15\%$	+8

Metric	Threshold	Points
Profit CAGR 3Y	< -10%	-6
EPS Growth	> 20%	+8
EPS Growth	< -10%	-8
Revenue acceleration	Latest > prior + 5%	+6
Revenue deceleration	Latest < prior - 5%	-6

Profitability

Metric	Threshold	Points
ROE	> 20%	+12
ROE	10-20%	+5
ROE	< 0%	-10
ROCE (screener)	> 20%	+10
ROCE	12-20%	+4
ROCE	< 6%	-6
Operating Margin	> 25%	+10
Operating Margin	10-25%	+4
Operating Margin	< 0%	-10
Net Margin	> 20%	+8
Net Margin	< 0%	-8

Cash Flow (yfinance + screener.in fallback)

Metric	Threshold	Points
Free Cash Flow	Positive	+10
FCF Growth	> 10% YoY	+6
Free Cash Flow	Negative	-8

Note: For Indian stocks where yfinance cashflow is empty, operating CF and investing CF are fetched from screener.in's #cash-flow section.

Leverage & Solvency

Metric	Threshold	Points
D/E	> 300%	-12
D/E	150-300%	-5
D/E	< 50%	+5

Metric	Threshold	Points
Current Ratio	> 2.0	+8
Current Ratio	1.2–2.0	+3
Current Ratio	< 1.0	–10
Net Cash Position	More cash than debt	+6

Shareholding (India)

Metric	Threshold	Points
FII + DII combined	> 50%	+6
FII + DII combined	25–50%	+3
Promoter holding	> 55%	+5
Promoter holding	< 25%	–5
Promoter pledge	> 50%	–15
Promoter pledge	25–50%	–8
Promoter pledge	10–25%	–3
Promoter pledge	0%	+3

7. Sentiment Analysis Module

File: `backend/services/news_sentiment.py`

Two-Layer Scoring

Layer 1 — Financial Lexicon (60% weight):

- ~80 domain-specific phrases with pre-calibrated scores
- Examples: "beat expectations" → +0.75, "profit warning" → –0.80, "upgrade" → +0.65
- Designed to override generic sentiment on financial language

Layer 2 — VADER Sentiment (40% weight):

- Title sentiment (70%) + Description sentiment (30%)
- Standard NLP library tuned for social media / news

Final blend: `score = 0.60 × lexicon + 0.40 × VADER`

News Sources (per prediction)

Source	Articles	Priority
Yahoo Finance RSS (<code>{symbol}</code> , <code>{symbol}INS</code>)	10	Primary
Google News RSS (<code>{symbol}</code> stock India)	10	Secondary
Economic Times RSS	10	India supplement

Source	Articles	Priority
MoneyControl RSS	10	India supplement

Cache TTL: 10 minutes per symbol

Sentiment Classification

Score	Label
≥ +0.05	BULLISH
−0.05 to +0.05	NEUTRAL
≤ −0.05	BEARISH

Score conversion: sentiment_score = 50 + (bullish − bearish) / total × 50

When no news available: Returns neutral (50), redistributes weight to technical + fundamental, sets data_available = False flag.

8. Global Macro Context Module

File: backend/services/global_context.py

Macro Indicators Tracked

Indicator	Ticker	What It Signals
S&P 500	^GSPC	US demand → IT/pharma export tailwind
NASDAQ	^IXIC	Tech hiring, demand for Indian IT services
VIX	^VIX	FII flows: high VIX → FII outflows from India
USD/INR	INR=X	Weak INR → pharma/IT revenue boost; import cost rise
Crude Brent	BZ=F	Oil import cost, OMC margins
Gold	GC=F	Jewelry demand (Titan, Muthoot)
Nifty IT Index	^CNXIT	Sector momentum for IT stocks
Nifty Bank Index	^NSEBANK	Banking sector rotation signal

Cache TTL: 15 minutes (fetched in parallel)

Stock-Specific Macro Sensitivity Map

Over 100 Nifty stocks mapped to their macro sensitivities:

Stock Category	Tailwind Factors	Headwind Factors
IT (TCS, INFY, WIPRO, HCL)	USD/INR weakness, S&P 500 up, NASDAQ up	INR strengthening
Pharma (SUNPHARMA, DRREDDY)	USD/INR weakness, S&P 500 up	—

Stock Category	Tailwind Factors	Headwind Factors
OMCs (BPCL, IOC)	Crude down	Crude up (margin squeeze)
Oil producers (ONGC)	Crude up	Crude down
Paints (ASIANPAINT, PIDILITIND)	—	Crude up (TiO2 input cost)
Banks (HDFCBANK, ICICIBANK)	—	VIX up (FII sensitivity)
Jewelry (TITAN)	Gold stable	Gold up (input cost)

Stock Adjustment Calculation

$\text{stock_adj} = \sum \text{sensitivity_i} \times (\text{factor_i} - \text{benchmark_i})$
 Tailwind factors: +2 to +4 points
 Headwind factors: -2 to -4 points

9. Quality Factors Module

File: backend/services/quality_factors.py

Applied for **India (Nifty 100)** only, on medium and long horizons.

10 Dimensions of Quality

1. Earnings Revisions (Weight: 12–14%)

- EPS surprise trend (beat vs miss last 4 quarters): ±16 pts
- Analyst upgrade/downgrade momentum: ±8 pts
- Forward PE compression vs trailing PE: ±8 pts

2. Institutional Ownership (Weight: 5–6%)

- Holdings > 50%: +14 pts
- Holdings 30–50%: +8 pts
- Holdings 15–30%: +3 pts
- Holdings < 5%: -5 pts
- Institution count > 300: +6 pts; < 20: -4 pts

3. Institutional Flow Proxy / MF Trend (Weight: 5–13%)

Blends price-volume signals (OBV trend, MFI, accumulation pattern) with real NSE FII/DII data:

- 60% price-volume proxy + 40% real FII/DII flows (when available)

4. Relative Strength (Weight: 7–15%)

Stock return vs NIFTY 50 over 1M, 3M, 6M:

- Outperform > 10%: +12 pts
- Outperform 4–10%: +6 pts
- Underperform > 10%: -12 pts
- Underperform 4–10%: -6 pts

5. Sector Strength (Weight: 7–15%)

Sector index momentum vs NIFTY 50:

- Sector outperform > 5%: +16 pts
- Sector outperform 2–5%: +8 pts
- Sector underperform > 5%: –14 pts
- Sector underperform 2–5%: –7 pts

6. Valuation Quality (Weight: 8–17%)

Multi-dimensional valuation scoring:

- PEG < 0.75: +16 pts; > 2.5: –12 pts
- EV/EBITDA vs sector: ±10 pts at major discount/premium
- Sector-relative PE: ±12 pts at 30% discount
- P/B for banks < 1.0: +12 pts; > 4.0: –8 pts
- Analyst target upside > 30%: +10 pts (margin of safety)

7. Risk Management (Weight: 10%)

- Max Drawdown 12M < –10%: +14 pts; < –40%: –14 pts
- Volatility Percentile bottom 25%: +10 pts; top 20%: –10 pts
- Sharpe Ratio > 1.5: +12 pts; < 0: –10 pts
- Downside Deviation < 10%: +6 pts; > 30%: –6 pts

8. Corporate Actions (Weight: 3–10%)

- Dividend payer 5Y+: +8 pts
- Growing dividends: +6 pts
- Payout ratio 0–40%: +4 pts; > 80%: –6 pts
- Active buyback: +8 pts
- Share dilution: –5 pts

9. Liquidity (Weight: 4–8%)

- Market cap ■2T+: +10 pts; < ■10B: –8 pts
- Avg daily volume > 5M: +8 pts; < 100K: –8 pts
- Beta 0.5–1.2: +4 pts; > 2.0: –5 pts

10. Quality Metrics (Weight: 4–12%)

Piotroski F-Score (9-point scale):

Point	Criterion
P1	ROA > 0
P2	Operating Cash Flow > 0
P3	ROA improving YoY
P4	Accruals: cash earnings > accrual earnings
P5	Leverage decreasing
P6	Current ratio improving
P7	No share dilution

Point	Criterion
P8	Gross margin expanding
P9	Asset turnover improving

- Score ≥ 8 : +20 pts
- Score 6–7: +10 pts
- Score 4–5: 0 pts
- Score < 4 : –12 pts

ROIC (Return on Invested Capital):

- $> 20\%$: +14 pts
- 12–20%: +8 pts
- 6–12%: +2 pts
- $< 6\%$: –8 pts

Horizon-Based Weighting

Dimension	Short	Medium	Long
Earnings Revisions	13%	14%	12%
Institutional Ownership	5%	6%	6%
MF/FII Flow	6%	8%	10%
Relative Strength	15%	11%	7%
Sector Strength	15%	11%	7%
Valuation Quality	8%	13%	17%
Risk Management	10%	10%	10%
Corporate Actions	3%	6%	10%
Liquidity	8%	6%	4%
Quality Metrics	4%	6%	12%
Flow Proxy	13%	9%	5%

10. Risk Penalty Framework

Applied **after** all signal scoring. Never adds points — only subtracts (risk override).

Risk Factor	Condition	Penalty
High leverage	D/E $> 300\%$	–8 pts
Elevated leverage	D/E 200–300%	–4 pts
High beta	Beta > 2.0	–6 pts
Elevated beta	Beta 1.6–2.0	–3 pts
Negative FCF	FCF < 0	–5 pts

Risk Factor	Condition	Penalty
Negative ROE	ROE < 0	–5 pts
Poor risk profile	Risk score < 35	–5 pts
Earnings volatility	CV > 0.5	–4 pts

Maximum penalty capped at –30 pts (prevents extreme scores on genuinely bad risk/reward stocks).

11. Confidence Engine

Answers: **"How much should you trust this signal?"**

Five Components

Component	What It Measures	Weight (Full)	Weight (Bootstrap)
Data Completeness	% of key fundamental fields present	25%	31.25%
Factor Agreement	% of factors agreeing with signal direction	25%	31.25%
Earnings Stability	Quality earnings_revision sub-score	15%	18.75%
Regime Certainty	Strength of current market trend (BULL/BEAR vs SIDEWAYS)	15%	18.75%
Historical Factor Reliability	Live IC values from outcome logit score (requires 60+ resolved predictions) (falls back to 50)	20%	10%

Sector-Aware Field Sets

Indian Non-Financial: PE, ROE, revenue growth, D/E, margin, beta (yfinance) + 3Y CAGR, ROCE, FII, promoter (screener) = 10 fields

Indian Financial (banks/NBFCs): PE, ROE, revenue growth, profit margin, EPS growth, beta (yfinance) + 3Y CAGR, FII, promoter (screener) = 10 fields

US Stocks: PE, ROE, revenue growth, D/E, margin, EPS growth, FCF, beta, ROCE = 13 fields

Confidence Bands

Score	Band
≥ 80	**HIGH**
60–79	**MEDIUM**
< 60	**LOW**

Percentile Context

Score	Label
≥ 80	Top 10% of all Nifty predictions
≥ 72	Top 20%

Score	Label
≥ 65	Top 35%
≥ 58	Top 50%
≥ 50	Below average
< 50	Low range

12. Target Price & Trade Levels

File: backend/services/prediction_engine.py → `_target_price()` , `_trade_levels()`

Target Price

Horizon	Method
Short	ATR × 2.5 × confidence factor — moves 1–5 day magnitude
Medium	Analyst mean target (70%) + price-based projection (30%); BUY floor: <code>`max(b</code>
Long	<code>`analyst_target × (1 + EPS_growth)^2`; BUY floor: <code>`max(target, price × 1.15)`</code></code>

Trade Levels

Take Profit = Model Target Price (consistent with signal)
 Stop Loss = ATR-based, adjusted to maintain minimum R:R of 1.5×
 Entry Zone = [price − 0.3×ATR, price + 0.1×ATR] for BUY
 [price − 0.1×ATR, price + 0.3×ATR] for SELL
 R:R Ratio = (target − price) / (price − stop_loss)

ATR multipliers by horizon:

Horizon	Base SL	Floor SL	Max SL
Short	1.5× ATR	1× ATR	25% of price
Medium	3× ATR	2× ATR	25% of price
Long	5× ATR	3× ATR	25% of price

13. Daily Picks Engine

File: backend/services/daily_picks.py

Execution Schedule

- Triggered: **Every weekday at 9:00 AM IST** (via GitHub Actions → POST `/api/picks/generate`)
- Generation time: ~10 minutes
- Results cached to disk: `backend/picks_cache.json`
- API response: Instant (reads from cache)

9-Phase Pipeline

Phase 0 — Outcome Resolution

- Compare previous predictions against actual forward returns (1-day / 63-day / 252-day)
- Update outcome logger database with direction hits (correct/incorrect)
- Feed data into IC engine for retraining

Phase 1 — Universe Screening

- Run full prediction engine on all Nifty 100 stocks
- Parallelised: 2 workers via `ThreadPoolExecutor`
- Returns raw factor scores for all stocks (enables cross-sectional z-scoring)

Phase 2 — Regime Detection

- KMeans clustering (4 clusters) on global macro features: VIX, S&P 500 return, crude, gold, USD/INR
- Classifies market into: **BULL_CALM**, **BULL_VOLATILE**, **BEAR_CALM**, **BEAR_PANIC**
- Returns regime label + weight multipliers for IC adjustment

Phase 3 — IC Weight Computation

- If < 60 outcome pairs: use academic prior weights
- Short: tech=0.055, fund=0.018, sentiment=0.042, quality=0.032
- If ≥ 60 pairs: Bayesian shrinkage blend of live IC + prior:
 $\text{weight} = \text{live_weight} \times \text{live_ic} + (1 - \text{live_weight}) \times \text{prior}$
- Apply regime multipliers: BULL boosts tech/sentiment; BEAR boosts fundamental/quality

Phase 4 — Z-Score Normalisation & Alpha

- Cross-sectional z-scoring: $z_i = (\text{score_i} - \text{mean}) / \text{std}$
- Combined alpha: $\sum \text{ic_weight_k} \times z_k$
- Meta-model alpha (requires 180+ outcomes across 60 stocks × 3 horizons):
- Inputs: tech_z, fund_z, sentiment_z, quality_z, combined_alpha, regime_id
- Output: predicted return %

Phase 5 — Pick Selection

- Rank by alpha score (meta_alpha if available, else combined_alpha)
- Select top 5 **BUY** signals per horizon
- Minimum 1 pick per horizon

Phase 6 — Portfolio Optimisation

- Fetch 6-month daily returns for selected picks
- Covariance estimation: Ledoit-Wolf shrinkage at 25%
- Optimise: $\max (\alpha \times w - \lambda \times w^T \Sigma w)$
- Constraints: $\sum w = 1.0$, $0 \leq w_i \leq 0.40$ (max 40% per position)
- Risk aversion λ : doubled in BEAR_PANIC regime

Phase 7 — Logging

- Log to PostgreSQL (if `USE_POSTGRES=1`) or SQLite
- Stored fields: factor z-scores, combined_alpha, meta_alpha, signal, price, horizon, regime

Phase 8 — Weight Adaptation (background)

- Retrain IC engine with new outcomes
- Update meta-model if sufficient data
- Recalibrate regime clustering
- Runs in daemon thread (non-blocking)

14. Backtesting & Validation Engine

File: `backend/services/validation_engine.py` , `backend/services/backtester.py`

Walk-Forward Methodology (No Look-Ahead Bias)

For each business day t in Nifty 100 history:

1. Fetch price data available **only before** time t
2. Compute prediction at t using that data
3. Measure actual forward return at $t + h$ ($h = 7 / 63 / 252$ days)
4. Compare predicted signal vs actual direction

Metrics Computed

Metric	Definition
Hit Rate	% of BUY/SELL calls that were directionally correct
Avg Return on BUY	Mean forward return when signal was BUY
Sharpe Ratio	$\frac{(\text{avg_return} - \text{risk_free})}{\text{std_return}}$ on BUY calls
vs Benchmark	Alpha over NIFTY 50 buy-and-hold
Score Calibration	Hit rate by score bucket (60–70, 70–80, 80–100)

Execution

- **Weekly:** Sunday 7:30 AM IST (via GitHub Actions)
- **Duration:** ~40 minutes (medium + long horizons)
- **Storage:** PostgreSQL `val_runs` + `val_signals` tables
- **Retention:** 365 days

Indicative Score Calibration (Nifty 100, Medium-Term)

Score Bucket	Hit Rate	Beat Benchmark	Avg Alpha
80–100 (Strong BUY)	~68%	~62%	+4.2%
70–79 (BUY)	~62%	~55%	+2.8%
60–69 (Moderate BUY)	~58%	~50%	+1.5%
55–59 (HOLD)	~52%	~48%	+0.3%

15. Crypto Prediction Module

File: backend/services/crypto_engine.py

Supported Assets

BTC, ETH, BNB, SOL, XRP, DOGE, ADA, AVAX, LINK, DOT

Signals Used (No Fundamentals)

Signal	Weight
Technical indicators (full suite)	Primary
Fear & Greed proxy (30-day vol / 90-day vol)	Secondary
On-chain proxy (price-volume accumulation)	Secondary
News sentiment (Google News + CoinTelegraph)	Supporting
Macro sensitivity (BTC/ETH correlated to S&P, VIX)	Adjustment

Quality factors and dynamic fundamental weights are not applied — crypto lacks P/E, ROE, cash flow data. Momentum and sentiment dominate.

16. Screener & Universe Management

File: backend/services/screener_service.py , screener_data.py

Screener Filters

Filter	Type	Description
Market	US / IN	Filter by market
Signal	BUY / HOLD / SELL	Filter by signal
Min Market Cap	■ Cr	Size filter
Max P/E	Ratio	Valuation filter
Min ROE	%	Profitability filter
Sector	IT / Banking / Pharma / etc.	Sector filter

Heatmap

- Groups stocks by sector
- Shows sector-level average return, top movers, stock count
- Colour-coded by performance (green/red intensity)

Top Movers

- Fetched in real-time from yfinance
- Ranked by absolute % change
- Separate for US, India, Crypto

17. API Reference

Prediction

Endpoint	Method	Description
<code>/api/predictions/{symbol}`</code>	GET	Get prediction (async, 202 while computing)
<code>/api/predictions/debug/state`</code>	GET	Internal cache/thread state (debug only)

Query params: `market` (US/IN/CRYPTO), `horizon` (short/medium/long)

Daily Picks

Endpoint	Method	Description
<code>/api/picks/daily`</code>	GET	Today's cached picks (instant)
<code>/api/picks/generate`</code>	POST	Trigger pick generation (secret-protected)

Screener

Endpoint	Method	Description
<code>/api/screener/filter`</code>	GET	Filter universe by fundamentals
<code>/api/screener/heatmap`</code>	GET	Sector heatmap
<code>/api/screener/top-movers`</code>	GET	Top 10 movers by % change
<code>/api/screener/crypto-movers`</code>	GET	Top 10 crypto movers

Backtest & Validation

Endpoint	Method	Description
<code>/api/backtest/run`</code>	POST	Single-stock walk-forward backtest
<code>/api/validation/run`</code>	POST	Full universe validation (background)
<code>/api/validation/status`</code>	GET	Validation progress
<code>/api/validation/results`</code>	GET	Validation metrics by horizon

Infrastructure

Endpoint	Method	Description
<code>/health`</code>	GET	Health check (keep-alive ping)

18. Frontend Pages & Components

Pages

Page	Route	Description
Dashboard	<code>`/`</code>	Top movers (US/IN/Crypto), market status, search
Stock Detail	<code>`/stock/:symbol`</code>	Full prediction, trade levels, factor breakdown
Daily Picks	<code>`/picks`</code>	Today's top BUY ideas by horizon + portfolio weights
Screener	<code>`/screener`</code>	Filter and explore the universe
Backtest	<code>`/backtest`</code>	Single-stock historical test
Watchlist	<code>`/watchlist`</code>	Saved stocks with live prices and signals
Alerts	<code>`/alerts`</code>	Price and signal alerts
Portfolio	<code>`/portfolio`</code>	Holdings with BUY/HOLD/SELL per position

Key Components

Component	Description
<code>`FactorAttributionWaterfall`</code>	Horizontal bar chart showing composite score decomposition; click any bar to
<code>`ConfidenceMeter`</code>	Colour-coded progress bar (0–100)
<code>`ConfidenceBreakdown`</code>	SVG gauge + 5-component confidence bars with tooltips
<code>`BullBearCase`</code>	Generated analyst-style bull/bear thesis bullets
<code>`TradingViewWidget`</code>	Embedded TradingView advanced chart (visual only; not connected to predict
<code>`IndexBar`</code>	Live NIFTY 50, SENSEX, VIX ticker strip
<code>`SignalBadge`</code>	BUY / HOLD / SELL badge with colour coding
<code>`ScoreHistoryChart`</code>	Composite score trend over time
<code>`NewsCard`</code>	Sentiment-tagged news article card

19. Infrastructure & Deployment

Hosting

Layer	Provider	Plan
Backend API	Render.com	Free tier (512 MB RAM, 0.1 vCPU)
Frontend	Vercel	Hobby
Database	PostgreSQL (Render) or SQLite (local)	—
Auth	Supabase	Free tier

Environment Variables

Variable	Description	Default
<code>`ENVIRONMENT`</code>	<code>`production`</code> / <code>`development`</code>	<code>`development`</code>
<code>`DATABASE_URL`</code>	PostgreSQL connection string	SQLite fallback
<code>`USE_POSTGRES`</code>	<code>`1`</code> = Postgres, <code>`0`</code> = SQLite	<code>`0`</code>
<code>`PICKS_SECRET`</code>	Secret header for <code>`/api/picks/generate`</code>	Required in prod
<code>`PICKS_UNIVERSE_LIMIT`</code>	Cap stock count (set < 25 on constrained hosts)	Full Nifty 100

Backend Startup Sequence

1. Uvicorn starts FastAPI app
2. CORS middleware enabled (frontend HTTPS origin)
3. Prediction router with daemon-thread async pattern initialised
4. Universe refresh (async, 30s delay)
5. Keepalive loop (14-minute interval)
6. Outcome resolver (6-hour interval)
7. Warmup loop — pre-computes RELIANCE:IN:medium and AAPL:US:medium (150s delay, 90s gap)

20. Automation Workflows

Directory: `.github/workflows/`

daily_picks.yml — Daily Picks Generation

```
Schedule: 0 3 * * 1-5 (3:00 AM UTC = 8:30 AM IST, Mon-Fri)
Action: POST /api/picks/generate with x-secret header
Purpose: Wake Render + trigger 10-min pick generation
Result: Picks ready by 9:00 AM IST
```

weekly_validation.yml — Model Accuracy Validation

```
Schedule: 0 2 * * 0 (2:00 AM UTC = 7:30 AM IST, Sunday)
Steps: 1. Ping /health to wake Render
2. POST /api/validation/run?horizon=medium → wait 15 min
3. POST /api/validation/run?horizon=long → wait 25 min
4. GET /api/validation/results → print summary
Output: BUY hit rate %, alpha %, Sharpe per horizon
```

keep_alive.yml — Render Free Tier Keep-Alive

```
Schedule: */10 * * * * (every 10 minutes, 24/7)
Action: GET /health
Purpose: Prevent Render free tier from spinning down
```

21. Factor Weights by Horizon

Short-Term (1–5 days)

Factor	Weight	Rationale
Technical	70%	Momentum and price action dominate short windows
Fundamental	15%	Slow-moving signal, limited short-term relevance
Sentiment	15%	News-driven price moves matter in 1–5 days
Quality	Not applied	Computation overhead; not predictive short-term

Medium-Term (2–4 weeks)

Factor	Weight	Rationale
Technical	40%	Trend still matters but less dominant
Fundamental	45%	Growth and valuation start to drive returns
Sentiment	15%	News flow still relevant over multi-week horizon
Quality	Included in composite	Institutional signals begin to matter

Long-Term (3–6 months)

Factor	Weight	Rationale
Technical	15%	Mean reversion reduces technical edge
Fundamental	75%	Business quality and valuation dominate
Sentiment	10%	Structural, not ephemeral, news matters
Quality	Included, weighted heavily	Piotroski, ROIC, cashflow most predictive

22. Key Design Principles

- No Look-Ahead Bias** — Backtester uses only data available at the prediction date. Forward returns computed strictly after prediction timestamp.
- Full Explainability** — Every signal has reasoning bullets, factor breakdown, and confidence components. Nothing is a black box.
- Honest Risk-Adjusted Returns** — Target prices are not inflated. Trade levels (entry, stop, target) are mutually consistent with the signal.
- Sector-Aware Scoring** — Different valuation thresholds for banks (ROE > 15%, P/B < 1.5), NBFCs, and IT companies vs industrials.
- Institutional-Grade Signals** — Piotroski F-Score, Sharpe Ratio, ROIC, IC engine, Ledoit-Wolf covariance optimisation — same tools used by quant funds.
- Data Resilience** — Three-layer fallback chain: yfinance → screener.in → BSE API. If news is unavailable, weights redistribute gracefully.
- Memory-Efficient** — Designed for Render's 512 MB free tier. Cache capped at 300 entries with LRU eviction. Concurrent predictions use daemon threads, not asyncio tasks.

8. **Self-Improving** — Outcome logger tracks every prediction. IC engine retrains weekly. Factor weights evolve as the model sees more real-world outcomes.
9. **Real-Time Ready** — 15-minute prediction cache, async background computation, React Query polling for live data.
10. **Investor Transparency** — Every number in the UI is traceable to a specific calculation in the codebase. This document is kept current with every code change.

This document is a living record of StockSense. It is updated with every significant change to the product.